

Lynbrook Math Summer Camp 2026

C3 Counting Techniques Part 2

Lynbrook Math Club

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C3.1 Lecture

Principle of Inclusion-Exclusion

You may be familiar with Venn Diagrams, and in this lesson we will use these concepts to solve harder problems.

Let's go over basic set theory notation and things that are important to know for this topic.

Important Terms and Symbols:

- **Set:** A collection of distinct objects
- **Union** ($A \cup B$): The set of all elements that belong to set A , set B , or both.
- **Intersection** ($A \cap B$): The set of elements that belong to **both** A and B . This is the overlapping region.
- **Cardinality** ($|A|$): The number of elements in set A .

When drawing the Venn Diagram for sets A and B , notice that the overlapping middle section ($A \cap B$) is part of circle A and also part of circle B . If you want to count how many total unique items are in either A or B , simply adding $|A| + |B|$ will count the middle overlap twice.

This leads to the **The Principle of Inclusion-Exclusion (PIE)**.

PIE can be used to count the number of elements in the union of multiple overlapping sets without double-counting.

Two-Set Formula: To fix the double-counting issue, we **include** everything in A and B , then **exclude** the duplicate counts in the intersection:

$$|A \cup B| = |A| + |B| - |A \cap B|$$

Three-Set Formula: What happens when three sets (A , B , and C) overlap?

1. **Include** all individual elements: $|A| + |B| + |C|$.
2. **Exclude** the pairwise overlaps: subtracting $|A \cap B| + |A \cap C| + |B \cap C|$.
3. **Correct for over-exclusion:** Notice that the center region where all three sets overlap ($A \cap B \cap C$) was added 3 times in step 1, but then subtracted 3 times in step 2. That means it isn't being counted at all. So we must include it back.

This gives the 3-set PIE formula:

$$|A \cup B \cup C| = |A| + |B| + |C| - (|A \cap B| + |A \cap C| + |B \cap C|) + |A \cap B \cap C|$$

For n sets:

$$\left| \bigcup_{i=1}^n A_i \right| = \sum_{1 \leq i \leq n} |A_i| - \sum_{1 \leq i < j \leq n} |A_i \cap A_j| + \sum_{1 \leq i < j < k \leq n} |A_i \cap A_j \cap A_k| - \cdots + (-1)^{n-1} |A_1 \cap A_2 \cap \cdots \cap A_n|.$$

For any number of sets, the pattern always alternates: **Add** individual sizes, **subtract** double overlaps, **add** triple overlaps, **subtract** quadruple overlaps, and so on.

Example 1 (AMC8 2015/15)

At Euler Middle School, 198 students voted on two issues in a school referendum with the following results: 149 voted in favor of the first issue and 119 voted in favor of the second issue. If there were exactly 29 students who voted against both issues, how many students voted in favor of both issues?

(A) 49 (B) 70 (C) 79 (D) 99 (E) 149

We can define our sets:

- A is the set of students who voted in favor of the first issue
- B is the set of students who voted in favor of the second issue
- $A \cap B$ is the set of students who voted in favor of both issues

We are given that:

- $|A| = 149$
- $|B| = 119$

Since 29 students voted against both issues, the number of students who voted for at least one issue is $198 - 29 = 169$.

PIE gives that $A \cup B = A + B - A \cap B$. Substituting our values in, we get

$$\begin{aligned} 169 &= 149 + 119 - A \cap B \\ \Rightarrow 169 &= 268 - A \cap B \\ \Rightarrow A \cap B &= 268 - 169 = \boxed{\text{(D) } 99}. \end{aligned}$$

Example 2

How many positive integers less than or equal to 100 are divisible by 2, 3, or 5?

Let A_2 , A_3 , and A_5 be the sets of numbers up to 100 divisible by 2, 3, and 5, respectively.

First, we count single set sizes using floor division $\lfloor \frac{100}{k} \rfloor$:

- $|A_2| = \lfloor \frac{100}{2} \rfloor = 50$

- $|A_3| = \lfloor \frac{100}{3} \rfloor = 33$
- $|A_5| = \lfloor \frac{100}{5} \rfloor = 20$

Next, we count pairwise intersections:

- $|A_2 \cap A_3| = \lfloor \frac{100}{\text{lcm}(2,3)} \rfloor = \lfloor \frac{100}{6} \rfloor = 16$
- $|A_2 \cap A_5| = \lfloor \frac{100}{\text{lcm}(2,5)} \rfloor = \lfloor \frac{100}{10} \rfloor = 10$
- $|A_3 \cap A_5| = \lfloor \frac{100}{\text{lcm}(3,5)} \rfloor = \lfloor \frac{100}{15} \rfloor = 6$

Then, we count triple intersection:

- $|A_2 \cap A_3 \cap A_5| = \lfloor \frac{100}{\text{lcm}(2,3,5)} \rfloor = \lfloor \frac{100}{30} \rfloor = 3$

Finally, we apply PIE:

$$|A_2 \cup A_3 \cup A_5| = (50 + 33 + 20) - (16 + 10 + 6) + 3 \\ \Rightarrow 103 - 32 + 3 = 74$$

There are 74 integers from 1 to 100 divisible by 2, 3, or 5.

Stars and Bars

Stars and Bars is a powerful counting technique useful for problems about dividing objects between groups. Suppose we have five identical stars.

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If we want to partition the stars among two people, we can put a bar in between the stars to put it into two groups. We could say that the stars to the left of the bar belong to Person 1 and the stars to the right of the bar belong to Person 2. Here are a few possible partitions:

★ ★ | ★ ★ ★ (2 to Person 1, 3 to Person 2)

★ ★ ★ | ★ ★ (3 to Person 1, 2 to Person 2)

★ ★ ★ ★ | ★ (4 to Person 1, 1 to Person 2)

| ★ ★ ★ ★ ★ (0 to Person 1, 5 to Person 2)

We see that finding the total number of partitions is equivalent to finding the total ways we can arrange the bars and stars, which we can find with combinations. There are 5 stars and 1 bar, so there are 6 items in total. We can choose 1 item to be the bar or choose 5 items to be the star, so there are $\binom{6}{1}$ arrangements in total.

If we extend this to partitioning n objects among k people, we would need $k - 1$ bars and n stars. Applying the same logic, there are $n + k$ objects, $k - 1$ of which have to be bars. Therefore, there are $\binom{n+k-1}{k-1}$ partitions.

Example 3

There are 5 students and 8 identical bags of chips. How many ways are there to distribute the chips if every student should get at least 1 bag?

Now, there is a restriction that every student must get at least one bag, so we can't directly apply our formula, because that may result in students getting no bags.

We can overcome this by giving 1 bag of chips to each of the 5 students in the beginning. Then there are 3 bags of chips remaining.

Now, we distribute the remaining 3 chips among the 5 students with no restrictions (students can now receive 0 additional chips).

Using the stars and bars formula with $n = 3$ remaining chips and $k = 5$ students we get that the number of distributions is

$$\binom{n+k-1}{k-1} = \binom{3+5-1}{5-1} = \binom{7}{4} = \boxed{35}.$$

Recursion

When we want to count how many ways there are to do a task with many steps, sometimes recursion can be a great tool to use.

For example, we may want to count how many ways there are to climb a staircase of certain steps where for each step we take, we can climb a different number of stairs. In this case, it can be easier to count how many ways there are to climb a 1-step staircase, a 2-step staircase, and so on until we can generalize it to n steps.

The steps we take when approaching a recursion problems are:

1. Define the sequence (a_n): Let a_n represent the total number of ways to solve the problem for size n .
2. Find the base cases: Solve the smallest, simplest cases manually (usually a_0 or a_1).
3. Establish the recurrence relation: Express a_n in terms of previous states like a_{n-1} or a_{n-2} .

Example 4

A frog is climbing a staircase with 6 steps. In each hop, the frog can jump up either 1 step or 2 steps. In how many different sequences of hops can the frog reach the 6th step?

First, we define a_n : Let a_n be the number of distinct ways to climb n steps.

Then, we need to find the recurrence relation. If we look the frog's last hop to reach step n :

- Option A: The frog came from step $n - 1$ via a 1-step hop. There are a_{n-1} ways to reach step $n - 1$
- Option B: The frog came from step $n - 2$ via a 2-step hop. There are a_{n-2} ways to reach step $n - 2$.

Because these two options cover all possibilities, so

$$a_n = a_{n-1} + a_{n-2}.$$

Now, to calculate a_n , we need to find the case cases.

- $a_1 = 1$ (There is only 1 way to go up 1 step, which is to hop 1 step)
- $a_2 = 2$ (The frog can jump up 1 step twice, or jump 2 steps)

Using these, we can calculate up to a_6 using our formula $a_n = a_{n-1} + a_{n-2}$:

- $a_3 = a_2 + a_1 = 2 + 1 = 3$
- $a_4 = a_3 + a_2 = 3 + 2 = 5$
- $a_5 = a_4 + a_3 = 5 + 3 = 8$
- $a_6 = a_5 + a_4 = 8 + 5 = 13$

Therefore, there are $\boxed{13}$ different hop sequences.

Distinguishability

Now that we've learned several techniques to count things, we can practice identifying which technique to use on different problems. In counting problems, whether objects are **distinguishable** or not is extremely important. If objects are distinguishable, they are different, and if they are indistinguishable, they are not.

Example 5 (AoPS)

How many ways are there to put 5 balls in 3 boxes if the balls are distinguishable but the boxes are not?

There are 5 different cases for arrangements of the number of balls in each box: $(5, 0, 0)$, $(4, 1, 0)$, $(3, 2, 0)$, $(3, 1, 1)$, and $(2, 2, 1)$.

- $(5, 0, 0)$: There is 1 way to put all the balls into 1 box.
- $(4, 1, 0)$: There are $\binom{5}{1} = 5$ ways to choose the ball in a box by itself.
- $(3, 2, 0)$: There are $\binom{5}{3} = 10$ ways to choose the balls in the group of 3 and 1 way to put the rest of the balls in a group of 2. $10 \times 1 = 10$ ways to arrange the balls.
- $(3, 1, 1)$: There are $\binom{5}{3} = 10$ ways to choose the balls in the group of 3, and 1 way to put the other two balls in separate boxes. There are $10 \times 1 = 10$ ways to arrange the balls.
- $(2, 2, 1)$: There are $\binom{5}{2} = 10$ ways to choose two of the balls to be together, $\binom{3}{2} = 3$ ways to choose another two to be together, and 1 way to put the last ball by itself in a box. However, we are overcounting here because the order of choosing the first pair and the second pair doesn't matter. Therefore, we must divide the total product by 2. Therefore, there are $\frac{10 \times 3 \times 1}{2} = 15$ ways.

Adding up the number of arrangements in each case, we get $1 + 5 + 10 + 10 + 15 = \boxed{41}$.

Example 6 (AoPS)

How many ways are there to put 5 balls in 3 boxes if the balls are indistinguishable but the boxes are distinguishable?

Since the balls are indistinguishable, we must only count the number of balls in the different boxes.

The ways to split the balls between boxes are $(5, 0, 0)$, $(4, 1, 0)$, $(3, 2, 0)$, $(3, 1, 1)$, and $(2, 2, 1)$.

- $(5, 0, 0)$: There are 3 ways to choose which box to put all the balls in.
- $(4, 1, 0)$: There are 3 ways to choose the box with 4 balls, 2 ways to choose the box with 1 ball, and 1 way to choose the box with no balls for a total of 6 ways.

- $(3, 2, 1)$: There are 3 ways to choose the box with 3 balls, 2 ways to choose the box with 2 ball, and 1 way to choose the box with 1 ball for a total of 6 ways.
- $(3, 1, 1)$: There are 3 ways to choose the box with three balls.
- $(2, 2, 1)$: There are 3 ways to choose the box with 1 ball.

Adding the number of ways in each case, we get $3 + 6 + 6 + 3 + 3 = \boxed{21}$ distributions.